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# Points to Consider in Remote and Hybrid GMP/GDP Audits

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## Authors


### Auditing Subteam

**Felicia Ford-Rice**, Parexel International (Subteam Lead)
**Robert L. Dana**, Elkhorn Associates, Inc.
**Gabriele Gori**, GlaxoSmithKline
**Alexander Nezlin**, UroGen Pharma
**Bryan Wilson**, GlaxoSmithKline

### Quality Risk Management Subteam

**Tiffany Baker**, ValSource, Inc. (Task Force Co-lead, Subteam Lead)
**Felicia Ford-Rice**, Parexel International
**Ghada Haddad**, PhD, Merck
**Eva M. Urban**, CSL Behring (Task Force Co-lead)

### Culture and Communication Subteam

**Karin Baer**, NeuroDerm (Subteam Lead)
**Vishal Sharma**, Vi'enni Training & Consulting
**Michele Simone**, Bracco
**Herbert Vail**, PQE Group
**Prashant Wange**, Sun Pharmaceuticals

### Technology Subteam

**Raghu Ram Pannala**, PhD, ScieGen Pharmaceuticals (Subteam Lead)
**William Cashin**, CordenPharma
**Siegfried Schmitt,** PhD, Parexel International
**Herbert Vail**, PQE Group

### Task Force

**Tiffany Baker**, ValSource, Inc. (Co-lead)
**Eva Urban**, CSL Behring (Co-lead)
**Karin Baer**, NeuroDerm
**William Cashin**, CordenPharma
**David Chesney**, DL Chesney Consulting
**Robert Dana**, Elkhorn Associates
**Alexandre Drapier**, GlaxoSmithKline
**Eser Earley**, Astellas
**Felicia Ford-Rice**, Parexel International
**Gabriele Gori**, GlaxoSmithKline
**Tor Gråberg**, AstraZeneca
**Ghada Haddad**, PhD, Merck
**Alexander Nezlin**, UroGen Pharma
**Raghu Ram Pannala**, PhD, ScieGen Pharmaceuticals
**Siegfried Schmitt**, PhD, Parexel International
**Vishal Sharma**, Vi'enni Training and Consulting
**Michele Simone**, Bracco
**Laura Singer**, Heron Therapeutics
**Prashant Wange**, Sun Pharmaceutical Industries Ltd.
**Herbert Vail**, PQE Group
**Bryan Wilson**, GlaxoSmithKline

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# Introduction

The ability of the pharmaceutical industry to conduct on-site audits has been impacted by global travel restrictions imposed during the global pandemic caused by the COVID-19 disease. These essential examinations are designed to ensure, through the detection of potential gaps, that all regulated systems are operating in a state of control or, if not, to aid in the timely detection and remediation of noncompliance. Industry has quickly pivoted to the use of remote auditing tools to ensure that the requirements of health authorities and marketing authorizations are being met.

This document describes steps that can be taken to ensure an effective and efficient remote assessment. It introduces standard nomenclature to describe the types of remote audits and discusses key considerations for their scheduling, duration, planning, conduct, closing, reporting, and follow-up. This document focuses on aspects of auditing that are different in the remote setting than in on-site audits or that have a particular impact in the remote setting. It may assist entities as they work through the challenges of remote and hybrid auditing and determine the changes in practices, standard operating procedures, and quality agreements that are necessary to facilitate effective and efficient remote audits.

In addition, **Appendix I** provides greater detail on how risk assessment principles can be applied to determine the format and frequency of remote audits. And **Appendix II** provides additional information that can be used to better understand the role of culture and communication during remote audits.

The Parenteral Drug Association Remote Inspections and Audits Task Force, which oversaw the development of this document, includes auditors, auditees, and consultants from around the world.

PDA developed this document with the goal of encouraging further dialog among industry, health authorities, and suppliers of technology and materials. It does not represent a standard or regulatory guidance.

## Scope

*PDA Points to Consider in Remote and Hybrid GMP/GDP Audits* applies to good manufacturing practice (GMP) and good distribution practice (GDP) audits of the following types:

- **Intercompany —** an audit of an external supplier that provides goods or services

- **Intracompany —** an audit of a manufacturing site by other staff of, or on behalf of, the same company that may include:
  - Internal and corporate audit, that is, an audit of functional departments, systems, documents, and processes to evaluate their state of control and compliance with corporate standards, and to drive process improvement. This may be conducted by locally based or global (corporate) quality assurance personnel from within the organization or by external experts
  - Self-inspection, that is, an audit of systems, utilities, documents, records, and processes to evaluate their state of control and compliance with applicable regulations, conducted in an independent and detailed manner by competent persons from the company or by external experts

This Points to Consider document provides information for both the auditor/sponsor/manufacturer and the auditee/supplier. The recommendations may be applicable to other types of GxP audits, although those were not specifically considered in the development of this document.

This document is not intended to address regulatory inspections, which are covered in the *PDA Points to Consider in Remote and Hybrid GMP/GDP Inspections*.

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# Glossary

### Audit
Process of conducting a systematic, impartial, and independent evaluation of systems and processes (including the collection of evidence) to determine whether they comply with established standards/requirements, are implemented effectively, and are suitable to achieve and maintain a state of control and to drive process improvement.

### Auditee
Unit, department, facility, site, or organization being audited.

### Auditor
Experienced and trained personnel responsible for planning and conducting audits, reporting, following up on audit reports, and escalating audit findings as necessary to reach resolution.

### Communication
A process by which information is exchanged between individuals through a common system of symbols, signs, or behavior; a verbal or written message ***(1)***.

### Culture
The customary beliefs, social forms, and material traits of a racial, religious, or social group and/or the characteristic features of everyday existence (such as diversions or a way of life) shared by people in a place or time ***(1)***.

### Desktop Audit
A type of remote audit in which documentation (e.g., site description and key procedures, drawings, quality records, and reports) is requested by the auditor who is not physically present at the site(s). Requests and responses to questions are shared in written form (e.g., via email or on a document-sharing platform). There can be several iterations of the requests and review of documentation, questions, and answers. Phone calls and videoconferences take place only rarely to provide clarification about the documents requested or provided.

### Document-sharing Platform
A secure software application that facilitates file transfer. Document-sharing platforms allow end-users to upload files to a shared storage space and designate who may access the files.

### Hybrid Audit
An audit that is conducted remotely in part and also includes an in-person presence and on-site visit by the auditor. The remote and on-site aspects may be concurrent or at different times (e.g., a portion of the audit is performed virtually or as a desktop audit, followed by a second portion on-site). The combination of remote and on-site components may be captured in a single audit report or in separate reports. This also may be referred to as a blended audit.

### Non-disclosure Agreement (NDA)
A legally enforceable contract that establishes confidentiality between the owner of protected information and the recipient of that information. This also may be referred to as a Confidentiality Agreement.

### Power Distance Index (PDI)
The extent to which the less powerful members of organizations and institutions (like the family) accept and expect that power is distributed unequally ***(2)***.

### Quality Agreement
A comprehensive written agreement between parties that defines and establishes each party's activities in terms of how each will comply with current good manufacturing practices. The quality agreement includes general parameters for audits.

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**Remote Audit**
An audit in which the auditor is not physically present at the site(s) where the activities subject to the assessment take place or have taken place. Remote audits may take several forms, including desktop, virtual, and hybrid.

**Technical Agreement**
A comprehensive written contract between parties involved in the contract manufacturing of a drug that defines and establishes each party's manufacturing activities. The agreement may cover general business terms and conditions such as confidentiality, pricing or cost issues, delivery terms, and limits on liability or damages. Technical agreements also may include elements of Quality Agreements related to responsibilities for good manufacturing practices.

**Supplier**
The entity with whom the agreement of the quality level is made and the transaction of purchase is made *(3)*. Suppliers may include contract manufacturers, contract testing facilities, distributors, warehousing facilities, other service providers, and providers of ingredients, raw materials, ancillary materials, and packaging components and services.

**Video Communication Platform**
Web-conferencing software that allows participants to conduct or attend meetings via the internet. This is also known as online meeting software or videoconferencing. The video communication platform enables remote meetings based on voice-over internet protocol (VoIP), online video, instant messaging, file-sharing, and screen-sharing.

**Virtual Audit**
In addition to sharing documentation, as in any other remote audit, virtual audits use technologies such as live or real-time streaming video, screen-sharing, or other means of real-time communication. Virtual audits may include virtual tours of aspects of the site being audited. These audits may also be referred to as distant audits or remote interactive evaluations.

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# I. Determining Audit Type

## Topic A.1: Using A Risk-Based Approach to Determine Audit Type

**Problem Statement**

What is the best way to determine the type of audit required (i.e., remote, hybrid, on-site)?

**Recommendation**

The auditor should use a risk-based approach to determine the appropriate type of audit (i.e., remote, hybrid, or on-site) based on the audit scope and objectives. This approach may be based on current safety and security needs (including the risk of travel) and the auditable entity's:

- Manufacturing type
- Regulatory compliance history
- Most recent audit history and ranking

An auditable entity may be an external supplier or an internal business unit.

Every audit involves a relationship between auditor and auditee. The auditor may need to consider aspects of this relationship in selecting the appropriate audit type. For instance, the auditee's willingness to share information in advance of an audit might impact the audit type.

**Appendix I** provides an example of a risk-based approach, including potential criteria and information on how to apply it.[1] It outlines recommendations for selecting the type of audit to be performed for:

- GMP auditable entities with whom the auditor has an existing relationship
- New auditable entities

**Rationale**

Using a risk-based approach allows the auditor to make consistent decisions regarding the type of audit required commensurate with the risk to product quality and patient safety.

[1] In any risk-based approach, each entity may consider additional risk factors and may weight the risks in a manner consistent with their own internal risk tolerance. PDA *Technical Report No. 54-6: Formalized Risk Assessment for Excipients*, for example, introduces risk classification for the product/services provided by a supplier along with risks associated with the quality target product profile, if applicable.

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# II. Scheduling a Remote or Hybrid Audit

## Topic A.1: Scheduling

**Problem Statement**

What are considerations for evaluation of audit schedules impacted by major disruptions to pharmaceutical manufacturing (e.g., COVID-19 travel restrictions, natural disaster)?

**Recommendation**

The auditor should use a risk-based approach to determine the frequency of the audit required based on the criticality of the department, intercompany site, supplier, system, process, product, and/or service.

The auditor should assess the maximum interval between audits that would allow demonstration of a state of control. Using risk assessment, the auditor may adjust the audit frequency, time frame, scope, elements, and/or areas to respond to developments that may impact the maintenance of the validated state. Such developments might include changes in key personnel, equipment, instruments, computerized systems, new procedures, or deficiencies resulting from complaints, health authority inspections, or a reduced workforce at the audited site.

Audit calendars are typically planned six months to one year in advance of an audit, and the dates are negotiated with the auditee. Auditors may need to plan remote audits further ahead to remain in compliance with company standard operating procedures (SOP) and policies, and to allow for flexibility with auditees that may not be able to accommodate the auditor's preferred schedule. For virtual audits planned well in advance, however, communication in the months leading up to the audit will be important to ensure that the site's processes and activities at the time of the audit are representative of the functions of interest to the auditor.

For audits of suppliers, the use of subcontracted audits or the purchase of independent third-party audit reports may be considered.

Where applicable, any required non-disclosure agreements and/or confidentiality agreements should be approved in advance of the audit.

Auditors should review SOPs and policies governing the conduct of audits to ensure that those documents allow for and describe the process for conducting remote audits.

**Rationale**

These considerations will allow for the planning and conduct of audits that may have been impacted by major disruptions to normal audit schedules and audit planning.


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## Topic A.2: Duration

**Problem Statement**

What are considerations to prevent virtual audits from extending beyond a reasonable period of time, especially across multiple time zones?

**Recommendation**

The duration of a virtual audit will be determined by multiple factors, including the following:

- Complexity of the audit
- Risk classification of the product or service
- History of use of the product or service
- Significance of any corrective actions
- Previous audit and compliance history
- Previous regulatory inspection history
- Terms of contract, quality agreement, and/or technical agreement
- Ease of access to and organization of documents, records, personnel, and facilities, including the auditor's ability to interview people working from home or at alternate sites and to monitor high-risk operations
- Format of data, documents, and records, i.e., electronic, paper, or hybrid
- Frequency and mode of communications
- Document-sharing platform selected
- Auditor's and auditee's availability and willingness to agree on audit duration, taking into account the potential for fatigue and the efficiency of the process
- Geographical location of auditee's facilities, time-zone differences between auditor and auditee, and language used by auditee personnel as compared to the auditor

To facilitate the efficient conduct of a virtual audit, best practices for an auditor include:

- Evaluating a predetermined statistical sampling of data, documents, and records
- Requesting key documents in advance of the audit to review prior to the audit start date
- Requesting key information in advance of the audit

**Rationale**

Consideration of these factors will allow auditors and auditees to determine and confirm a reasonable virtual audit duration and ensure that the audit process will be effective and efficient.

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# III. Planning a Remote or Hybrid Audit

## Topic A.1: Planning the Agenda for a Virtual or Hybrid Audit

### Problem Statement

Typically, the audit agenda includes systems and or processes to be covered and the dates of the audit. How will this be impacted in a virtual or hybrid audit?

### Recommendation

When performing virtual or hybrid audits, certain elements must be considered:

- The virtual audit agenda should be more specific and detailed than an on-site audit, and it should include not only systems, but also specific topics, dates, and times to ensure the auditee can determine the availability of the appropriate experts. In planning the agenda, the auditor and auditee also should agree on the timing of the closing meeting, which indicates the formal end of the audit.
- The agenda should specify the key personnel from each party who will participate in the opening meeting, discussions, daily wrap-up meetings, and audit closing meeting.
- The schedule should allow additional time for potential issues with technology or personnel availability.
- In the virtual and hybrid format, the auditor should carefully consider the priority of different aspects of the audit. The auditor should clearly identify the audit topics and/or systems that are crucial for assessment to ensure that appropriate time is allotted. While important in any audit, this prioritization is especially important in the virtual and hybrid settings because of the potential for technological constraints as well as time limitations.
- If live translation will be used for discussions, the audit schedule should include extra time.
- The audit agenda should include reasonable breaks in the discussion so that the auditor and auditee can refresh themselves.
- Multiple workstreams may be required, for instance, when different auditors are reviewing different material in a single audit. In these circumstances, the agenda should provide for parallel sessions.
- Reasonable accommodations should be made to ensure that the audit is conducted within hours that are manageable for both parties, especially when the auditee's personnel and facilities are in significantly different time zones than the auditor. Third-party audits are scheduled during the auditee's normal working hours.

The auditor and auditee should agree on the agenda well in advance of the audit, so that all parties can make the necessary preparations.

### Rationale

An accurate and preplanned agenda for virtual audits that follows the recommendations above will ensure that the overall audit process is effective and efficient. Virtual audits are inherently less flexible than on-site audits due to the time and challenges involved in exchanging information.


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## Topic B.1: Planning the Technology

### Problem Statement

What technology is typically used during the conduct of a remote audit? What agreements need to be in place for the use of technology?

### Recommendation

While virtual audits, by definition, require the use of technology, each audit may use different technologies. Remote audits can use the following methods, individually or in combination:

- Teleconference—real-time dialog between the auditor and the auditee by telephone
- Video communication platform—real-time dialog through a platform that allows the auditor and auditee to see one another and share presentation materials
- Video—live or recorded, showing the auditee's facility and equipment
- Email correspondence
- Read-only database access—the auditor is provided access to the auditee's system
- Document-sharing platform—the auditor is provided with read-only access to the auditee's documentation and records
- Data exchange—access to data is provided on proprietary servers

Consideration should be given to the following:

- Availability of appropriate infrastructure for the audit (e.g., software, licenses, internet bandwidth) for both the auditee and the auditor. In a virtual or hybrid audit, the video communication platform and the document-sharing platform have a significant impact on the flow and the ease of the audit. Software that is frustrating for auditors to use may result in a longer-than-anticipated audit.
- The need for both parties to have robust and efficient technical support available
- Whether either party would be allowed to record data, documents, personnel, or facilities, including screen shots of virtual proceedings, document repository information, and audio or video of livestreamed discussions. This discussion should include any necessary review and approval by legal counsel for both parties.
- Identification of the areas of the facility that should be made available through photos or through live or prerecorded video
- Cybersecurity precautions that may prevent data exchange and access to data and document repositories, including firewalls, incompatible software, or information technology (IT) restrictions imposed by the auditee or auditor
- Cybersecurity risks that may arise for both parties when the auditee grants access to its IT systems
- For a virtual audit, alternative modes of continuing an audit in the event of environmental issues that cause loss or fluctuation of power, equipment failure, or loss of Wi-Fi
- A plan to respond to disruptions due to emergencies, loud sounds, loss of video, or lack of availability of key personnel during a virtual audit
- The battery life of equipment that will be used to conduct virtual tours. If batteries will require charging, this should be addressed in the agenda.

Best practices include:

- Establishing an agreement in advance between the auditor and the auditee about the technology to be used, which can prevent delays at the start of the audit
- Defining the document-sharing platform, which may be a file-sharing service or cloud-based service
- Determining whether the auditee is able to provide information about the technological options and any limitations
- Testing all technology in advance of a virtual audit, including the document-sharing platform, video communication platform, Wi-Fi network (for robustness), and connectivity of the devices

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in all areas of the site and remote locations. Determine whether all participants for both auditor and auditee are competent in the technology before the audit begins. Any issues identified during testing, including any limitations on software use based on geographical region, elapsed time, and number of users, should be resolved before the audit begins.

- Defining the minimum internet speed required for all participants in a virtual audit, to ensure that communications are clear and uninterrupted (e.g., 10 Mbps upload speed, where possible), with a robust connection and consistent communication throughout the audit process
- Agreeing on alternative communication options in case of a communication system failure during a virtual audit
- Addressing access restrictions due to legal concerns, safety, Wi-Fi, internet access, controlled classified areas, or other reasons, and agreeing on the areas that may and may not be viewed through video
- Determining legal requirements regarding personal information, including restrictions on showing faces of personnel in a virtual audit

**Rationale**

Selection of the proper technology platforms that consider both security and data privacy is important for information exchange and for the development of trust between auditors and auditees. Prior agreement on the technology to be used will help assure a successful audit by allowing the process to be more efficient and robust. Establishing an agreement in advance also allows both parties to install and test any necessary technology before the audit begins, which will prevent delays during the audit itself.


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## Topic C.1: Planning for Document-Sharing

**Problem Statement**

What is the best approach for ensuring that critical documents and records are available and understandable during a remote audit?

**Recommendation**

The scope and types of documents to be shared in a remote audit, along with the appropriate formats, any restrictions, and period of time they will be available for review, should be defined by the two parties. For initial remote and hybrid audits, this information may be described in a document that reflects the agreement between the parties (e.g., mutually agreed-upon audit checklist or agenda). To plan for the future, however, the quality agreement between the parties should be revised to address these scenarios.

Before a virtual or hybrid audit begins, the two parties should agree on the documentation that will be shared in advance of the audit, along with timelines for sharing, and the documentation that will be shared only during the audit. Sharing some documentation before the start of a virtual or hybrid audit can help prevent delays while the auditor becomes familiar with the information and formulates questions. Furthermore, the quality of the audit may be compromised if the auditee refuses to share any documents in advance. This, in turn, may impact the relationship between auditor and auditee. The agreement also may define a time frame after which the auditee will disable the auditor's access or delete the documents.

The auditee should allocate time for document preparation and organization, including the appropriate formatting of documents for uploading and sharing. The auditee also should develop a strategy for determining the number and types of data, documents, and records that will need to be accessed and evaluated.

The auditor should discuss with the auditee the language of the auditee's documents and records, without assuming that the documents and records are in the auditor's native language. If the documents are not in the auditor's native language, the parties should agree on whether the auditee can provide translated documents and records, or if a translator must be made available. If a translator is necessary, a neutral third-party translator should be used.

In preparation for a remote audit, the auditee should prepare documents in digital format. Documents might be provided in a variety of ways, e.g., through a secure portal, by providing a limited-access database, through video, or using other technological platforms. If mutually agreed for a virtual or hybrid audit, the auditee may share discrete, specific information from a hard-copy document (e.g., one page from a logbook) using a document camera, which is a real-time image-capture device that functions as a high-resolution web camera, mounted on an arm for stability, with magnification, if necessary. The parties should limit the use of email for delivery of documents and records due to firewalls, security measures, and limitations in the size of email attachments.

The auditee should ensure that the electronic files are accessible by the auditor. The auditee can send files in a digital form that can be opened without requiring any specific software application. For example, documents in PDF or CSV format can be opened using many different types of software. Common formats for data include word processing documents, spreadsheets, and video. Other software that supports publishing over the internet also may be useful.

The auditor and the auditee should discuss and agree on the best way to organize documents and records for ease of retrieval. The auditee also should communicate clearly to the auditor the location of requested data, documents, and records, once shared.

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The auditee should ensure that the full title, version, and effective date of documents is clearly available so the auditor can capture this information for the audit report.

Before sharing materials, the auditee should assess the materials for consistency, accuracy, completeness, auditability, orderliness, uniqueness, and timeliness. In a remote audit, these issues are more difficult to remediate after providing the materials to the auditor; therefore, an additional control check may be necessary in the remote setting. During the audit, the auditee should keep the physical documents on site in a well-organized manner in case it is necessary to cross-check or share the document in a different format.

It is highly recommended that the auditee provide documents in a format that is searchable by the auditor. To accomplish this, any scanning should be done in optical character recognition (OCR) format.

To protect confidential information, the use of encryption technology should be considered. In general, where possible, the auditee should place restrictions on all documents shared, including disabling screen-copying and restricting printing and copying.

If site personnel will need to access site-specific software during the audit, they should verify in advance that the software is accessible from the computer system that will be used during the audit. Software specific to the manufacturing location or laboratory, including software used for issuance of batch records, temperature and pressure monitoring, and other integrated manufacturing software systems, may not be accessible from all company computers or may reside on networks that are decoupled from the internet. Advance preparation will ensure that personnel can participate from the manufacturing location or laboratory or can take other necessary steps.

Auditors and auditees should update their SOPs to allow for remote and hybrid audits and to reflect their current approaches to the elements discussed in this section and throughout this document.

**Table 1** presents considerations about specific types of data that may be shared as a part of a virtual or hybrid audit.

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**Table 1** Options for Sharing Specific Types of Data in a Virtual Audit

| Type of Information | Technology Option | Advance-Planning Considerations |
|---|---|---|
| Standard operating procedures (SOPs) and related documents | Share through document-sharing platform. | SOP indices may be shared before the audit, which will help in reducing audit time. |
| Critical quality attribute (CQA) and critical process parameter (CPP) data | The auditee can navigate through specific software during the remote audit through screen-sharing during the web conference. In general, the auditee should not grant the auditor access to or direct control of the auditee's systems to avoid cybersecurity threats as well as inadvertent user error. | CQA and CPP data will be shared only as mutually agreed. |
| Quality metrics data such as out-of-specification log, deviations log, out-of-trend log, complaints log, change control log | If manual logs are maintained, the logs can be scanned or can be displayed live using a document camera.<br>If quality assurance management system software is used, the software can be displayed through screen-sharing. | It may not be necessary to share quality metrics in advance of the audit. |
| Building management data, area qualification data, and environmental data | The most recent periodic environmental monitoring report can be scanned in OCR format or livestreamed using a document camera.<br>Other supporting information would be provided during the course of the audit using screen-sharing or a document camera. | |
| Equipment calibration plan and conclusions | Livestream through a document camera. | In advance of the audit, identify the equipment used in the lab and in manufacturing for the products that are or may be within the audit scope. |
| Laboratory information management system (LIMS) database | Be prepared to share a report or validation, along with audit trail, by screen-sharing if requested. | It may be useful to verify in advance that LIMS software and electronic notebooks are easily retrievable from archives. |
| Laboratory and chromatographic software | Share via screen-sharing. Be prepared to display the raw data, metadata, and audit trail through screen-sharing as well, if requested. | |
| Manufacturing software | Specific records of the batches within audit scope can be extracted into PDF format along with the audit trail in preparation for the audit.<br>The software may be demonstrated through videos or shared through screen-sharing. Be prepared to display the raw data, metadata, and audit trail through screen-sharing. | |
| Packaging software, track-and-trace/ serialization software | Share via screen-sharing. Be prepared to display the raw data, metadata, and audit trail through screen-sharing. | |

## Rationale

The auditor and auditee should agree upon approaches that best ensure that critical documents and records are available, viewable, and secure to allow for an effective compliance assessment.

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## Topic C.2: Preventing Communication Challenges During Document-Sharing

### Problem Statement

What are approaches for overcoming communication challenges during the sharing of documents?

### Recommendation

In a remote audit, each party understands that the auditee will upload or share large quantities of documents with the auditor. This activity may present several challenges that can be addressed with proper planning, including the development of a document-sharing strategy, and clear communication.

For instance, one challenge likely to arise is unfamiliarity with the document-sharing platform, which may generate discomfort and/or apprehension for both auditor and auditee. Another challenge for both the auditor and the auditee is the difficulty in predicting the amount of time that will be needed to prepare for and conduct the remote audit. This uncertainty can present a major motivational challenge. For example, it may be difficult to predict:

- For the auditee, the time necessary to identify, scan, demarcate, and classify documents, and ensure the completeness of all uploads
- For the auditor, the time necessary to confirm the receipt of all necessary documents against a checklist and/or quality questionnaire and to understand the organization of the documents and the activities they describe
- For both parties, the time necessary to become familiar with the software platform, conduct any necessary training on the platform, and troubleshoot any technical issues
- For both parties, the time necessary to successfully complete the audit at the appropriate depth

To proactively prevent such challenges from developing, the auditee should develop a document-sharing strategy. Document-sharing is one of the early steps of remote audits, and a professionally executed document-sharing process leaves a positive impression on the auditor. The strategy should focus on identifying the needs of the auditor, which may be divided into four parts ***(4)*:**

1. Baseline expectations—considered a norm
2. Spoken expectations—expressed explicitly and above the baseline expectations
3. Unspoken expectations—normally not expressed until violated
4. Delighters—steps the auditor may desire but does not expect to be provided, though providing them creates a strong positive impact. The auditor may not see these as true "needs."

The document-sharing strategy formulated by the auditee should consider the steps or actions that would meet the auditor's baseline expectations, as well as the additional steps that would meet the auditor's normal expectations, which may be spoken or upspoken. However, the auditee may also consider the actions that would exceed the auditor's expectations and create a "delight" by easing the auditor's navigation through the documents and supporting the auditor's efforts to understand the documents. **Table 2** provides an example of how the auditee might classify such expectations.

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**Table 2** Example for Identifying Auditor's Needs for Developing Document-Sharing Strategy for Remote Audits

| Type of Expectation | Example of Auditor's Need | Example of Auditee's Action to Satisfy Auditor's Need |
|---|---|---|
| Baseline Expectation | Auditee will share all relevant data and documents per the scope of audit | Seek clarity on the scope of audit; get an agreement (verbal and written) on the type of documents to be shared and a list |
| | | Have auditors confirm receipt of the documents |
| Normal Quality – Spoken Expectations | Relevant documents will be uploaded two weeks prior to the date of the audit | Confirm in advance the list of documents that must be provided |
| | | If more time is needed to prepare and share the documents, communicate with auditors before the deadline the reason for the request for extension; provide facts, if necessary, and set a realistic new timeline |
| Normal Quality – Unspoken Expectations | Navigation of the documents and trails is effortless, and documents are easy to access, read, and understand without assistance | Help the auditor become familiar with the document-sharing platform; provide clear, detailed login information and appropriate technical support |
| | | Upload a presentation or guide with the documents to explain their location in the shared folder |
| | | The presentation/guide should also briefly explain the content of the documents and their relationship with the other documents |
| Delighters – Exceed Expectations | Process of unassisted reading is intuitive | Before introducing the auditor to the document-sharing platform, perform a simulation of the auditor's experience in which a first-time user at the auditee site acts as the auditor and experiences the platform navigation and document-reading unassisted; use this simulation to identify pain points and improve the process before sharing documents with the auditors |

## Rationale

Proactive development of a document-sharing strategy, with training and support for personnel executing the document-sharing strategy, can help ensure efficient audits and can significantly reduce stress. A single document-sharing strategy can be used across audits and can be modified with new learnings after implementation.

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# IV. Conducting a Remote or Hybrid Audit

## Topic A.1: Opening Meetings

### Problem Statement

How can opening meetings be modified to best suit the virtual nature of the audit?

### Recommendation

Opening meetings for virtual audits should be structured to ensure efficiency by considering the following aspects:

- Ensure that all personnel participating in the audit are announced by title and function for transparency. A graphic may be provided showing the name, title, and photo of audit participants as a substitute for the exchange of physical business cards.
- Schedule time specifically for introductions and pleasantries for everyone on the call.
- Take time to walk through and agree on the audit agenda to ensure that necessary subject matter experts, documents, and records are available and to identify and address any obstacles or technical issues. Ensure that the subject matter experts will be available at the times described in the agenda and make any necessary adjustments.
- Identify the technologies that the parties have agreed to use during the audit, as described in **Topic III.B.1 Planning the Technology** and **Topic III.C.1 Planning for Document-Sharing**. Address any questions about how documents will be accessed, how personal interviews will be conducted, and how the facility tour (if any) will be conducted.
- Discuss the ground rules of the day's session and how it is expected to proceed to demonstrate that everyone's time and expertise is valued.
- If not already provided for in the agenda, establish the timing of breaks during the day, as well as the daily wrap-up meeting.
- Make it clear that sipping beverages as necessary is acceptable and appropriate.
- Allow that information provided in advance may not need to be presented live at the opening meeting.
- Provide presentations in advance of the audit, so the audit may move directly to asking questions and holding meaningful discussions.
- Avoid having auditee personnel read from prepared slides, if presentations are made; instead, provide an overview of the company and function.
- Consider alternative formats for the facility overview, in lieu of or in combination with an opening PowerPoint presentation. For instance, prerecord videos or use livestream video, if agreed, as described in **Topic III.B.1 Planning the Technology.**
- Provide a floor plan that can help orient auditors when reviewing video footage, allowing them a better understanding of the locations shown in live or prerecorded video.
- Confirm how to communicate throughout the audit (e.g., use of group chat, email, text).
- Reiterate the contingency plans for technological or other issues (e.g., connection issues, loss of power) and acknowledge challenges that may arise as a result of working remotely.

### Rationale

Opening meetings may involve a large number of auditee attendees and therefore may require additional time for introductions. Virtual audits should be structured to ensure that all attendees, including senior management in various geographic locations, have an opportunity to introduce their roles in the company. With time constraints and the auditor's lack of physical access to the site, the measures noted above will help ensure that the opening meeting is effective and efficient.


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## Topic B.1: Communication

**Problem Statement**

What are some means of ensuring effective verbal and nonverbal communication during remote audits?

**Recommendation**

Remote audit preparation activities should include training or awareness of verbal and nonverbal communication styles to avoid misunderstanding, especially across cultures. This should include understanding the impact of interpersonal communication. Personnel that will be involved in the audit should be made aware of culture-specific communications etiquette.

The parties should discuss in advance whether translation will be needed during live discussions and, if so, decide which party will provide the translator. A translator that is a native speaker of the auditor's language can help reduce stress relating to communications.

Verbal communication skills and awareness for teleconferences and videoconferences include such tips as:

- **Speaking Volume and Enunciation:** Ensure that each participant can hear and be heard clearly. The clarity of speech may be impacted by the technology that is used during the audit. Speakers that are required to wear a face mask due to shared working spaces should take particular care to be sure that they are clearly understood.

- **Speed of Talking:** Speaking speed should be moderated during remote audits, for the easy comprehension of listeners in other countries. A neutral speaking speed averages 100 to 150 words per minute, which may feel significantly slower than normal speaking speed. If a live interpreter is involved in the conversation, further moderation of speaking speed may be necessary. Research on the speaking rates of broadcasters, highlighted in **Table 3**, shows that speaking rates can vary dramatically, even between speakers of the same language in different regions ***(5-6)***.

**Table 3** Average Number of Words Spoken per Minute by Broadcasters

| | U.S. (CNN) | UK (BBC) | France (Radio France) | Italy (RAI) | Spain (RNE) |
|---|---|---|---|---|---|
| Words Per Minute | 146 | 167 | 189 | 192 | 210 |

- **Duration of Talking Time:** In advance of the audit, participants should be provided training or be made aware of the appropriate length of speaking time. The auditee should avoid providing responses that are monologues (at one end of the spectrum) or monosyllables (at the other end). The average duration of talk and the length of sentences is impacted by cultural and other factors. Cultural factors also may impact the listener's motivation and etiquette in interrupting speakers. This may be influenced by the power distance index (PDI) of the organization, which is discussed further in **Appendix II**.

- **Interruptions:** In the virtual environment, interruptions can be especially noticeable. Auditors and auditees should give careful attention to the topic of interruptions and interjections, as interruptions during discussions could be viewed as inappropriate, rude, or confrontational, even if they are unintentional.

- **Acknowledgement versus Understanding:** It is important for all personnel involved in the audit to clearly understand the difference between acknowledgement of a request or question and understanding of the request or question. It can be useful to ask frequently whether the question and answer are clear.

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- **Silence:** Silence can indicate several different things—that the person is taking time to think through a question and response, did not hear or does not understand the question, or perhaps does not know how to respond.

- **Mute Etiquette:** Video communication platforms generally allow users to mute their own microphones, and for the "organizer" or administrator of the meeting to mute a participant's microphone. The organizer should provide notice before muting another participant.

- **Topic Transitions and Clarification of Topics:** Because the auditee may not always know what documents the auditor is reviewing or looking at, the auditor needs to provide context for questions when changing the topic or referring to a different document. Without these transitions from one topic to another, confusion and delays may occur as the auditee reorients to the auditor's new topic.

- **Camera Etiquette:** In virtual and hybrid audits, the parties should agree early about whether and in what circumstances the participants will turn off their cameras and participate through audio only. This will help avoid confusion and miscommunication.

- **Avoiding Distractions:** All participants should disable device notifications during the audit to limit distractions and interruptions.

**Appendix II** includes discussion of one method that can be used to understand cultural differences between societies. It provides specific steps that can be taken to sensitize personnel to differences in verbal communications that may result from these cultural differences.

While a remote audit poses challenges in verbal communication, nonverbal communication remains the most challenging. The teams on both sides of the computer screens are at a disadvantage in that they cannot read the body language of the other party. For instance, it may not be possible to know whether a pause by the auditor or auditee is deliberate or is a technical glitch.

To prepare for online communication during virtual audits, the auditor's and the auditee's organizations should devote additional work to several aspects of nonverbal communication, as some communication experts believe nonverbal communication forms two-thirds of communications *(7)*. Nonverbal communication skills and awareness preparation for videoconferences should include such points as:

- Facial expressions are important to intensify, downplay, hide, or mask feelings.
- Prolonged eye contact can convey having a superior or dominant position, as well as being truthful and credible.
- Avoidance of eye contact can indicate shyness, lack of attention, guilt, or lack of knowledge. Lack of eye contact also may reflect cultural norms. Some auditees may not look directly at the computer's camera.
- Posture can be interpreted to convey low or high energy, confidence, and the person's general attitude, e.g., whether the person is interested, defensive, engaged, confident, impatient, or angry.
- Touch can be an indicator of nervousness, especially touching one's face.
- Gestures, especially when made in the presence of those from different cultures, should be controlled.

A communication plan for virtual audits will provide tools to preempt the challenges presented by nonverbal communication cues and to neutralize communication challenges. **Table 4** includes points to consider while devising the appropriate strategy.

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**Table 4** Considerations for Nonverbal Communication in Virtual Audits

| Nonverbal Communication Aspect | On-Site Audits | Virtual Audits |
|---|---|---|
| **First Impression** | A first impression is a lasting nonverbal communication ***(8)***.<br>According to a study from Princeton University, it takes one-tenth of a second for someone to judge another and form their first impression. Business attire is a standard that has evolved over time to provide a neutral yet professional impression during on-site audits. | A virtual audit, relying upon on-camera discussions, poses additional challenges beyond the attire worn during the audit. The camera angle and distance between the face and the camera can provide a neutral and professional impression during the in-camera audits. |
| **Posture** | Posture is a nonverbal cue used to determine a participant's degree of attention or involvement. In on-site audits, a mirror-image congruent posture, where one person's left side is parallel to the other person's right side, leads to favorable perception of communicators and positive speech ***(9)***. | During virtual audits, mirror-image congruent postures may require an understanding of the jaw thrust, shoulders, and arm movement. Touching one's face or crossing arms during on-camera discussions may provide a much stronger visual impression than the same actions performed during on-site audits. |
| **Gestures** | Gestures may be made with the hands, arms, or body and include movements of the head, face, and eyes. Different cultures have specific gestures to communicate specific meanings, e.g., a hand gesture to communicate good-bye. However, some near-universal gestures provide similar meaning across cultures, e.g., a shoulder shrug. During on-site audits, tightening of jaw muscles, furrowing of forehead, squinting of eyes, or occlusion of lips convey that the communicator is experiencing tension ***(10)***. | During virtual audits, gestures play a much more prominent role than on-site audits, as the virtual platform increases the effect of the gesture. While jaw tightening, forehead furrowing, eye squinting, or lip occlusion will communicate tension, so will the oscillating movement of a revolving chair, touching the eyeglasses, or frequent adjustment of the camera. |
| **Conversational Gestures** | Conversational gestures are hand movements that accompany speech and are related to the speech they accompany, e.g., pointing fingers at a particular object while providing directions to someone. During on-site audits, neutrality in communication is achieved by reducing the speed and distance of hand movements. | In virtual audits, it is important to restrict conversational hand movements completely to achieve neutrality in communication. Frequent conversational gestures, e.g., bringing one's ear closer to the laptop in an attempt to hear better, can have a jarring effect on the viewer, in part due to delays in image transmission from variable internet speed. |
| **Science of Proxemics** | The amount of space between two people engaged in communication displays the proximity and level of informality between the two ***(11)***. During on-site audits, neutrality of communication may be achieved by keeping at least an arm's-length distance between an auditor and auditee of two different cultural backgrounds. | During virtual audits, the person's distance from the camera and the amount of background area visible displays the proximity and degree of informality. Neutrality of communication may be achieved by keeping one's frame in a white or neutral background. |
| **Eye Contact** | Eye contact is the instance when two people look at each other's eyes at the same time; it is the primary nonverbal way of indicating engagement, interest, attention, and involvement ***(12)***. During on-site audits, constant eye contact between the auditor and auditee is used to communicate attentive listening and interest. | During virtual audits, a participant may need to view documents projected on the screen or to read from written documents available at hand. Frequent eye contact with the computer's camera is helpful in depicting interest. Continuous looking at the camera without shifting one's gaze to other objects depicts active listening. |
| **Use of Time** | The way time is used can send nonverbal messages. During on-site audits, auditors measure the time it takes the auditee to present a requested document to judge the aptness of a documentation system for filing and retrieval. | During virtual audits, records are either presented to the auditor in advance or are shared live. Even a time lag of a few seconds while searching for a document in the shared folder or navigating to a particular section of a document may impact the auditor's opinion and impression. |

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**Rationale**

Establishing trust between the auditor and auditee is of paramount importance in any audit and can be more difficult in the remote setting. Verbal and nonverbal communications play an important role in building trust. Building internal awareness of relevant communication strategies in advance of remote audits will help participants identify and overcome these challenges.


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# V. Wrap-up and Closing of a Remote or Hybrid Audit

## Topic A.1: Daily Wrap-Up and Audit Closing Meetings

**Problem Statement**

What is the most effective means of conducting a daily wrap-up meeting for remote audits? What should be taken into account for the remote audit closing meeting?

**Recommendation**

The use of daily wrap-up meetings and end-of-audit closing meetings are best practices in conducting virtual and hybrid audits and are essential to effective virtual auditing. These meetings allow for (1) understanding of the auditor's needs to access personnel, records, and facilities; (2) auditee preparedness for the upcoming day; and (3) inclusion of key personnel. As virtual and hybrid audits can involve personnel across multiple time zones, these meetings are important to ensuring remote audit efficiencies as well as the timeliness of the remote audit conduct.

For desktop audits, a closing meeting usually is not necessary, but the auditor should clearly communicate that the audit is complete.

The following should be considered in planning and scheduling a **daily wrap-up meeting** in the virtual context:

***Scheduling***

- Meetings should be concise, short in duration, and scheduled to follow the completion of the daily audit.
- The timing for the wrap-up meetings should be agreed by the parties. If conducted on the same day, a break between the end of the audit day and the meeting should be scheduled to allow the auditor to review notes and prepare observations.
- As an alternative, meetings could be scheduled for the beginning of the next audit day.

***Content***

- Meetings should reiterate the findings shared during the day.
- The auditor should provide verbal feedback on positive aspects of the audit, and preliminary and substantive observations.
- The auditor should state the criticality of observations, if known.
- The auditor should provide a high-level overview of any compliance issues identified, with specific checks for understanding.
- Both parties should discuss technical or logistical challenges. Discuss the effectiveness of the different technologies used during the day, highlight what worked and what didn't, and make any necessary changes for the next day.
- The meeting should include discussion of what the auditor needs, so the auditee can prepare for the next audit day.
- The auditor should provide clarity on what topics or areas are considered complete, and which will require additional time for evaluation, including virtual tours, personnel interviews, or records review.

A closing meeting should be scheduled following the completion of the audit. The following should be considered in planning and scheduling an **audit closing meeting**:

***Scheduling***

- A break should be scheduled between the end of the final audit day and the meeting to allow the auditor to review notes and prepare preliminary observations.
- As an alternative, the meeting could be scheduled for the following day.

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***Content***

As a good practice, the auditor should thank the auditee for their time and support.

The auditor may choose to display a visual of audit findings to the auditee (e.g., PowerPoint presentation), discuss audit findings and feedback verbally, or use a combination of both.

The auditor should discuss:

- Positive aspects of the audit
- Compliance issues and risks
- Areas evaluated and records and documentation assessed
- Limitations and challenges encountered during the audit
- Classification of observations as preliminary since an internal review typically is required prior to issuing observations
- Next steps for written communication of audit findings

The auditee should be given the opportunity to ask questions to ensure mutual understanding and to request clarification of findings.

**Rationale**

In the daily wrap-up meetings and audit-closing meeting for virtual audits, the auditor should present all significant concerns raised during the course of the day or audit. Both auditor and auditee should communicate understanding of the follow-up information required for assessment, key personnel to interview, whether additional correspondence is acceptable and, if so, within what time frame, and any additional evaluation. The audit closing meeting should be used to communicate significant compliance risk and expected next steps for the audit reporting process.


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# VI. Reporting in a Remote or Hybrid Audit

## Topic A.1: Documenting the Nature of an Audit as Remote or Hybrid

**Problem Statement**

How should an auditor document that an audit was remote or hybrid? What are considerations for ensuring traceability of the nature of the audit as remote or hybrid across the auditing process?

**Recommendation**

Audit reports and any corresponding audit certificates should include a reference to the nature of the audit (e.g., desktop, virtual, or hybrid) and to the technologies used in its conduct. The report also should indicate any topics and areas that could not be covered due to the type of audit performed and any requested information that was not provided. For example, if an area of the facility was not accessible due to the nature of a remote audit, or if no video or photos of live operations were shared, this should be noted in the report. Similarly, if documents were only shared during the actual audit, but not before or after the audit dates, this also should be noted for future reference, as this may have impeded the auditor's ability to perform the audit.

The report should document whether or not the technology used was controlled by the auditee. If the auditee declined to use specific technology that was requested (e.g., use of video for the purpose of virtual tour of facilities), this should also be documented in the report.

The auditor also should consider whether the nature of the audit (e.g., desktop, virtual, or hybrid) is recorded in the system in which the audit report is archived. Where possible, the audit report, audit certificate templates, and/or electronic systems used to generate audit reports should be designed to make it easy to both (1) record (e.g., during audit scheduling, audit report authoring, and archiving) whether the audit was desktop, virtual, or hybrid, and (2) specifically record the information necessary to allow future searches by the nature of the audit conducted.

**Rationale**

Including details about the nature for the audit will create an historical record of the approach used to conduct the audit, any difficulties encountered with the technology, and how these difficulties were addressed. This information will also be useful in conducting a comparison of the compliance history of the auditee against any desktop, virtual or hybrid audit challenges recorded.

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# VII. Follow-up to a Remote or Hybrid Audit

## Topic A.1: Follow-Up

**Problem Statement**

Are there best practices for following up after remote and hybrid audits?

**Recommendation**

The follow-up for remote and hybrid audits is similar to the follow-up for on-site audits.

The auditor should consider whether the auditee may require additional time to provide evidence of correction of observations or findings.

More time may be necessary to confirm the correction of observations or findings once on-site audits can be conducted; this should be taken into consideration when scheduling a follow-up on-site audit.

If a follow-up audit must be conducted remotely, the auditor and auditee should reach agreement regarding the objective evidence that would be acceptable to confirm compliance with regulatory requirements.

Trend analysis on the frequency and/or category of observations may be helpful to determine whether auditors are primarily focusing on certain aspects, and perhaps limiting their focus on other areas during virtual audits, as compared to how they would conduct on-site audits.

**Rationale**

These steps will help ensure that the auditor's and auditee's expectations relating to remote audit follow-up are communicated, agreed upon, and do not adversely affect the assessment of compliance.


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# IX. Relevant Global Guidances


## United States

- U.S. Food and Drug Administration, *Guidance for Industry: Remote Interactive Evaluations of Drug Manufacturing and Bioresearch Monitoring Facilities During the COVID-19 Public Health Emergency*, April 2021, https://www.fda.gov/media/147582/download (last accessed 07June2021)

- *Coronavirus (COVID-19) Update: FDA Updates on Surveillance Inspections During COVID-19*, 11 May 2020, https://www.fda.gov/news-events/press-announcements/coronavirus-covid-19-update-fda-updates-surveillance-inspections-during-covid-19 (Last accessed 08Dec2020)

- *Coronavirus Disease 2019 (COVID-19) Update: Foreign Inspections*, 10 March 2020, https://www.fda.gov/news-events/press-announcements/coronavirus-disease-2019-covid-19-update-foreign-inspections (Last accessed 08Dec2020)

- *Guidance for Industry: Manufacturing, Supply Chain, and Drug and Biological Product Inspections During COVID-19*, *Public Health Emergency Questions and Answers*, May 2021, https://www.fda.gov/regulatory-information/search-fda-guidance-documents/manufacturing-supply-chain-and-drug-and-biological-product-inspections-during-covid-19-public-health (Last accessed 19Nov2021)

- *Guidance for Industry: Resuming Normal Drug and Biologics Manufacturing Operations During the COVID-19 Public Health Emergency*, September 2020, https://www.fda.gov/regulatory-information/search-fda-guidance-documents/resuming-normal-drug-and-biologics-manufacturing-operations-during-covid-19-public-health-emergency (Last accessed 08Dec2020)

- *MDSAP Transmittal Number: 2020-07*, 03Jun2020, https://www.fda.gov/media/136441/download (Last accessed 08Dec2020)

## Australia

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## Europe

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# X. Appendix I: Determining Audit Type Using a Risk-Based Approach

## Topic A.1 Audit Type: New Relationship Auditable Entity

### Problem Statement

What risk-based approach can be used for new auditable entities with no previous or existing relationship?

### Recommendation

The proposed risk-based model presented in this section considers two criteria when selecting the appropriate audit type for a new auditable entity (e.g., new supplier, testing lab, contract manufacturing organization).[1] The two criteria for this assessment are the criticality of the services provided by the auditee and the regulatory inspection history.

### *Preliminary Assessment*

Before proceeding to consider the two criteria, the auditor must determine whether the risk-based approach is appropriate in light of the proposed scope and intent of the audit by using the filtering decision tree illustrated in **Figure AI.1-1**. The auditor should consider the following question for each auditable entity: "Does the audit scope require physical examination of the facility?" A new relationship with a contract manufacturer, for example, usually would require on-site examination, whereas the auditor may determine that an on-site evaluation is not required for a new relationship with a software or metrology vendor.

If on-site presence is not required, the auditor can use the risk-based approach described below to determine audit format.

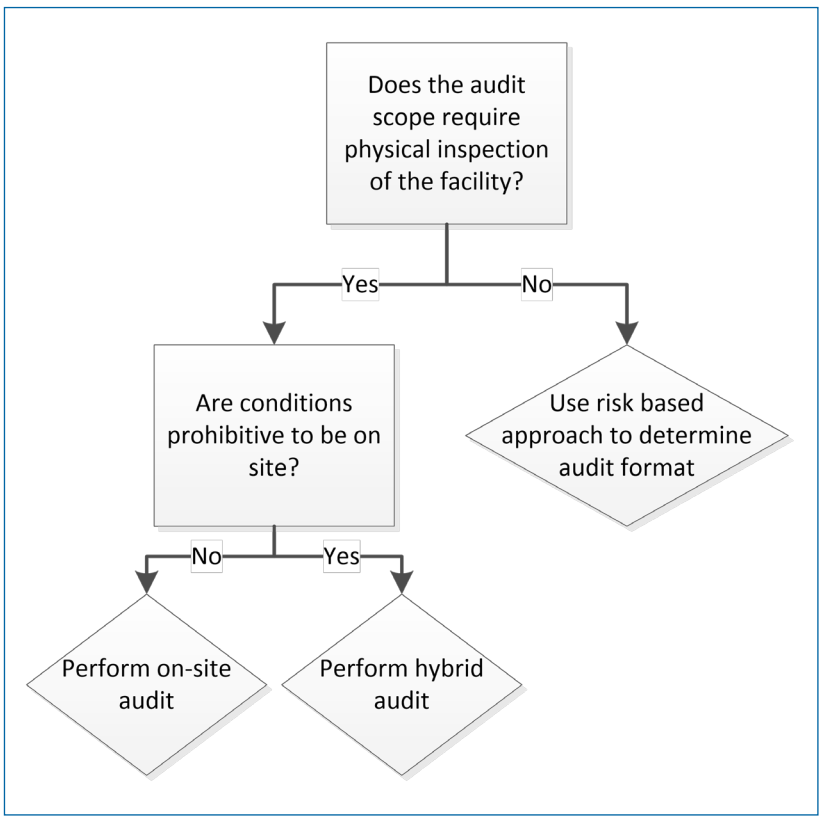


**Figure AI.1-1** Filtering Decision Tree

[1] In any risk-based approach, each entity may consider additional risk factors and may weight the risks in a manner consistent with their own internal risk tolerance.

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If the audit will require on-site examination, however, the auditor would then consider a second question: "Are conditions prohibitive to be on site?" If conditions are prohibitive (e.g., the Covid-19 pandemic did not allow for travel), a hybrid approach could be considered using technology to aid the assessment as defined in **Topic B.1**. This hybrid approach could allow for remote sections to occur while conditions are prohibitive, with necessary on-site portions to occur once travel can resume.

### *Risk-Based Tool*

If the entity is a candidate for the risk-based approach, the auditor would proceed by considering the two criteria that are essential to the risk-based determination of audit format. The first criterion involves the categorization of the auditable entity based on the criticality of the services it provides. The criticality of the services provided increases as the activities approach direct product contact. **Table AI.1-1** provides examples.

**Table AI.1-1** Criticality of Services Provided Criteria

| Criteria Rank | Criticality of Services Provided Criteria† |
|---|---|
| Critical | Services provided have direct impact on product quality or compliance. Sterile manufacturing, sterile product, primary packaging, medical devices (inclusive of combination products), biological drug substance (API), cell banking, sterile product testing laboratory, HVAC certification services |
| Moderate | Services provided have indirect impact on product quality or compliance.<br>API, API intermediate, nonsterile primary packaging, clinical packaging, laboratory (finished-product testing), on-site formulation (nonsterile facility), nonsterile manufacturing facilities |
| Minor | Services provided do not have impact on product quality or compliance.<br>Laboratory in-process testing (specialty testing), secondary packaging, warehouse logistics, document storage companies, distribution centers, warehousing companies |

† The criticality of a supplier that is the only available source of the material (sole source) can be determined based on the auditor's risk tolerance.

The second criterion is based on the auditable entity's current[2] regulatory compliance history (see **Table AI.1-2**). The better the auditable entity's current regulatory standing, the lower the risk ranking.

In the case of an auditable entity that has undergone recent regulatory inspections by several different inspectorates, the most conservative (i.e., highest) score should be applied. For suppliers of excipients, which may not be inspected by regulators, independent GMP certification may be used to verify compliance status.[3]

**Table AI.1-2** Regulatory Compliance History Criteria and Risk Ranking

| Risk Rank | Auditable Entity Regulatory Compliance History Criteria |
|---|---|
| High | Recent regulatory inspection yielded regulatory action (e.g., import alert, import ban, GMP certificates revoked, warning letter), or no regulatory history exists for organization |
| Medium | Health Authority findings (e.g., FDA 483) or equivalent categorization findings identified through recent inspections, but no further regulatory action expected |
| Low | Recent regulatory inspection with no findings or minor findings |

The risk matrix shown in **Figure AI.1-2** can be used to determine the risk based on the auditable entity's current regulatory compliance history and the criticality of the services provided.

[2] Current regulatory compliance period can be set according to the company's risk tolerance (e.g., previous one to three years)

[3] PDA *Technical Report No. 54-6: Formalized Risk Assessment for Excipients* may be useful in further defining the risks presented by the supplier.

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| Criticality of Services Provided | Regulatory Compliance History: Low | Medium | High |
|---|---|---|---|
| Critical | Medium | High | High |
| Moderate | Medium | Medium | High |
| Minor | Low | Low | Medium |

**Figure AI.1-2** Risk Level Matrix (Criticality of Services Provided × Regulatory Compliance History)

**Example**: **Table AI.1-3** illustrates the risk assessment of two auditable entities based on the model proposed in this section, resulting in a risk score for each.

Company A, a supplier responsible for sterile manufacturing of a component that has had a recent regulatory inspection with no findings, would be scored as Medium risk.

Company B, a finished product contract testing laboratory that has no regulatory inspection history, would be scored as High risk.

**Table AI.1-3** Example Risk Assessment Worksheet for New Auditable Entity

| Auditable Entity | Mfg. Type Criticality | Regulatory Compliance History | Risk Score | Justification |
|---|---|---|---|---|
| Company A | Critical | Low | **Medium** | Company A is a sterile manufacturer = Critical rating for criticality of services provided<br>Recent regulatory inspections resulted in no findings = Low rating for regulatory compliance history |
| Company B | Moderate | High | **High** | Company B is a contract testing lab = Moderate rating for criticality of services provided<br>No regulatory inspection history exists = High rating for lack of regulatory compliance history |

The action level table (**Table AI.1-4**) defines the recommended audit type based on the auditable entity's overall risk score. This action level table is designed to ensure adequate control where necessary.

A remote audit is acceptable in the event of a low-risk determination. A hybrid audit is recommended if the risk-based model determines that the auditable entity's risk is Medium; however, a remote audit can be acceptable if the appropriate justification is provided (e.g., review of past audit history, CAPA responses, or determination that performing a remote audit is better than performing no audit). If the risk is determined to be High, however, an on-site audit would be recommended.

In some instances, on-site audits will not be possible (e.g., due to COVID-19-related travel restrictions or safety protocols, or inability of auditable entity to host an audit). If an on-site audit is recommended but cannot be conducted, additional controls and more thorough hybrid audit techniques may be appropriate. The requirements of all relevant health authorities, based on market authorization holder filings, should be considered when determining the audit format.


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**Table AI.1-4** Action Level based on Risk Score

| Risk Score | Action Level |
|---|---|
| **High** | On-site audit recommended; external auditable entity should not be used if remote is the only audit option available |
| **Medium** | Hybrid audit recommended; remote audit possible with justification |
| **Low** | Remote audit is acceptable |

These risk-based approaches should not be construed as applying solely to the determination of audit type. These approaches may also be used to assist with other risk-based decisions, such as audit frequency or scope determination.

## Rationale

Using a risk-based approach allows consistent audit planning commensurate with the risk to product quality.


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## Topic A.2: Audit Type: Existing Relationship with Auditable Entity

### Problem Statement

What risk-based approach can be used to determine audit type for an auditable entity with which the auditor's organization has a previous or existing relationship?

### Recommendation

In this model, the auditor uses a filtering decision tree and three criteria to determine the appropriate audit type for an auditable entity with which the auditor's organization has an existing relationship. For this assessment, the auditor will utilize two matrices to arrive at the final risk score.

As a first step, the auditor will use the same filtering decision tree described in **Topic A.1** and illustrated in **Figure AI.1-1** to determine whether the risk-based tool is appropriate.

If the entity is a candidate for the risk-based approach, the auditor would consider the three criteria that are essential to the risk-based determination: criticality of the services provided, historical audit performance, and operational performance.

The first criterion is the criticality of the services provided by the auditable entity. **Table AI.2-1** provides an example of the classification of auditable entities based on the authors' experiences, although it is not designed to be comprehensive. Criticality of the services provided increases as the auditable entity's services approach direct product contact.

**Table AI.2-1** Risk-Ranking based on Criticality of Services Provided

| Risk Rank | Criticality of Services Provided Criteria† |
|---|---|
| Critical | Services provided have direct impact on product quality or compliance. Sterile manufacturing, sterile product, primary packaging, medical devices (inclusive of combination products), biological drug substance (API), cell banking, sterile product testing laboratory, HVAC certification services |
| Moderate | Services provided have indirect impact on product quality or compliance.<br>API, API intermediate, nonsterile primary packaging, clinical packaging, laboratory (finished-product testing), on-site formulation (nonsterile facility), nonsterile manufacturing facilities |
| Minor | Services provided do not have impact on product quality or compliance.<br>Laboratory in-process testing (specialty testing), secondary packaging, warehouse logistics, document storage companies, distribution centers, warehousing companies |

† The criticality of a supplier that is the only available source of the material (sole-source) can be determined based on the auditor's risk tolerance.

The second criterion is the historical auditing performance of the auditable entity. The period considered for the historical review can vary. For example, an auditor may look solely at the most recent audit or may look at all audits within a certain time window. A standard review period can be established based on the auditor's risk tolerance, for example, one year, two years, or three years.

**Table AI.2-2** provides examples of how to apply the historical audit performance criteria. If there were no findings within the selected time period, the auditable entity's risk ranking is Low. One major or several minor findings results in a Medium risk-ranking, and critical findings would result in a rating of High risk for this element.


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**Table AI.2-2** Historical Audit Performance Criteria and Risk-Ranking

| Risk Rank | Historical Audit Performance Criteria |
|---|---|
| High | Audit(s) resulted in at least one critical finding, or the content of the findings potentially reflect a high compliance risk related to product quality and quality systems, or a regulatory action may result if not corrected. |
| Medium | Audit(s) resulted in a total of three or more major and minor findings, but no critical findings.<br>If there were any repeat findings, a higher risk ranking may be considered. |
| Low | Audit(s) resulted in only minor findings or suggested improvements that do not impact quality or compliance of the site. No repeat findings from most recent audit. |

The auditable entity's criticality of services provided is then assessed against the historical audit performance to obtain a preliminary risk level. **Figure AI.1-3** displays the preliminary risk-level matrix based on these two criteria. This score will be factored into the final matrix (**Figure AI.1-4**).

The third criterion is the auditable entity's operational performance (**Table AI.2-3**). Operational performance is an indicator of the auditee's relationship with the auditor as well as the history of response with the auditor's organization. The better the communication and response to past audit findings, the lower the risk rating for this criterion. If operational performance data is not available, the preliminary matrix can serve as the final risk level until such data is available. The collection and documentation of this category of information may need to be implemented at an organization level before it can be consistently applied across all suppliers.

**Table AI.2-3** Auditable Entity Operational Performance Criteria

| Risk Rank | Auditable Entity Operational Performance Criteria |
|---|---|
| Poor | Auditable entity has not responded to findings adequately or in a timely manner following previously conducted audits. Lack of trust exists between auditor and auditable entity; communication is often delayed and is not transparent. Auditable entity has history of goods failing to meet quality specifications, has a history of repeated Supplier Corrective Action Requests (SCAR), or routinely fails to deliver goods on time. |
| Moderate | Auditable entity responds to findings from previously conducted audits, but plan requires discussion and revision with auditor. Communication with auditable entity occurs freely but is often delayed, some trust exists between entities. Auditable entity has received a SCAR in the auditor's routine look-back period. |
| Good | Auditable entity provides adequate response plan to findings from previously conducted audits. Well established communication with auditable entity exists that is transparent and timely. Auditable entity has demonstrated history of goods meeting quality specifications and routinely provides goods on time. |

The final risk matrix to determine the type of audit needed, shown in **Figure AI.1-4**, combines the output from **Figure AI.1-3**, Preliminary Risk Level Matrix, with the Operational Performance ranking from **Table AI.2-3**.

| | | Historical Audit Performance† | | |
|---|---|---|---|---|
| | | Low | Medium | High |
| Criticality of Services Provided | Critical | L2* | L3 | L3 |
| | Moderate | L1 | L2 | L3 |
| | Minor | L1 | L1 | L2 |

*L = Preliminary Risk Level as determined by Manufacturing Type x Historical Audit Performance

† This table does not use red, yellow, and green zones because the results are placeholder values that will carry over and be combined with the next criterion to obtain the overall risk level.

**Figure AI.2-3** Preliminary Risk Level Matrix (Criticality of Services Provided × Historical Audit Performance)

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| Preliminary Risk Level | Operational Performance Rating: Good | Moderate | Poor |
|---|---|---|---|
| L3 | Medium | High | High |
| L2 | Low | Medium | High |
| L1 | Low | Low | Medium |

**Figure AI.2-4** Risk Level Matrix (Preliminary Risk Level × Operational Performance Rating)

## Examples

Company C, a contract testing laboratory responsible for conducting sterility testing, has had a recent audit with critical findings. They are open and transparent in the relationship with the auditor's organization and timely in responses.

Company D, a backup contract testing laboratory, has had a recent audit with no critical but three major findings. They have historically required revisions to audit responses but have an overall good relationship with the auditor's organization.

Company E, a supplier responsible for sterile manufacturing of a component, has had a recent audit with critical findings. There is a lack of trust with this auditee, a tendency to provide inadequate responses or, in some cases, a failure to respond at all.

**Table AI.2-4** presents a worksheet of the risk-based assessment for Companies C, D, and E.

**Table AI.2-4** Example Risk Assessment Worksheet for Auditable Entities with Existing Relationship

| Auditable Entity | Services Provided Criticality | Historical Audit Performance | Preliminary Risk Level | Operational Performance | Overall Risk Score | Justification |
|---|---|---|---|---|---|---|
| Company C | Critical | High | L3 | Good | Medium | Company C is a contract testing laboratory responsible for conducting sterility testing = Critical rating for services provided<br>Recent audit history includes critical findings = High rating for audit performance<br>Transparent, open, and timely communication = Good operational performance |
| Company D | Minor | Medium | L1 | Moderate | Low | Company D is a backup contract testing laboratory = Minor rating for services provided<br>Recent audit history includes no critical but some major findings = Medium rating for audit performance<br>Responses to past findings often requires revision = Moderate operational performance |
| Company E | Critical | High | L3 | Poor | High | Company E is a supplier responsible for sterile manufacturing of a component = Critical rating for services provided<br>Recent audit history includes critical findings = High rating for audit performance<br>Lack of trust in auditee, late or absent communication, tendency to provide inadequate responses to past findings = Poor operational performance |


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The action-level table, **Table AI.2-5,** can be used to translate the level of risk determined through this approach to the most appropriate type of audit to be performed when there is an existing relationship between auditor and auditable entity. A remote audit is acceptable in the event of a Low-risk determination. A hybrid audit is recommended if the risk of the auditable entity is Medium; however, a remote audit can be acceptable if an appropriate justification is provided. If the risk is determined to be High, however, an on-site audit is recommended.

These risk-based approaches should not be construed as applying solely to the determination of audit type. These approaches may also be used to assist with other risk-based decisions, such as audit frequency or scope determination.

**Table AI.2-5** Action Level based on Overall Risk Score for Auditable Entity with Existing Relationship

| Overall Risk Score | Action Level |
|---|---|
| **High** | On-site audit recommended; hybrid audit possible with justification |
| **Medium** | Hybrid audit recommended; remote audit possible with justification |
| **Low** | Remote audit is acceptable |

**Example**: **Table AI.2-6** identifies the action level and audit type recommendations for Companies C, D, and E, based on the assessments in **Table AI.2-4**.

**Table AI.2-6** Action Level Ranking for Example of Auditable Entities with Existing Relationship

| Auditable Entity | Overall Risk Score | Audit Type Recommended |
|---|---|---|
| Company C | **Medium** | Hybrid audit recommended; remote audit possible with justification |
| Company D | **Low** | Remote audit is acceptable |
| Company E | **High** | On-site audit recommended; hybrid audit possible with justification |

In some instances, on-site audits are not possible due to COVID-19-related travel restrictions. If the auditor determines, through use of a risk-based approach, that an on-site audit is recommended but is not possible, additional risk-based controls and more thorough hybrid audit techniques may be considered.

## Rationale

Using a risk-based approach allows consistent audit planning commensurate with the risk to product quality.


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# XI. Appendix II: Additional Considerations for Understanding the Role of Culture and Communication in Remote Audits

## Topic A.1: Use of Hofstede's Power Distance Index to Create an Effective Verbal Communication Strategy for Remote Audit

**Problem Statement**

How does the Power Distance Index (PDI) impact communications in a remote audit? Using the PDI, what steps can be taken to sensitize personnel to communication differences when preparing for an audit?

**Recommendation**

The change from on-site to remote audits can alter long-standing communication patterns. Cultural differences in communication and behavioral patterns can become more apparent. Cultural clues that were implicitly acknowledged in on-site audits, and often considered to be "non-issues," may be more difficult to read and interpret in remote audit settings. These differences may have an impact on the seamless preparation and execution of the remote audits.

The auditor and auditee teams should devise a verbal communication plan for virtual audits, given the complexity of the language, its interpretation, and the cultural dimensions. To estimate the different efforts required, both the auditor and auditee should use all available tools to understand the significant aspects of one another's organizational culture and communication patterns. This will help create an efficient communication strategy and help ensure a successful remote audit.

***Background on Hofstede's Cultural Dimensions Theory and the Power Distance Index***

One tool available to train audit staff to be aware of geographical cultural differences and to assist in preparing a communication plan is Hofstede's Cultural Dimensions Theory. Hofstede's theory discusses the effects of an organization's culture on the behavior of its members and on cross-cultural communications. It uses a framework derived from factor analysis ***(1)***. All six dimensions of culture, as established by Hofstede, are relevant to the pharmaceutical industry and should be considered:

1. Individualism versus Collectivism
2. Masculinity versus Femininity
3. Uncertainty Avoidance Index
4. Long-term Orientation versus Short-term Normative Orientation
5. Indulgence versus Restraint
6. Power Distance Index

Of these, the PDI is especially relevant to consider in the change from on-site to remote audits. While the other cultural dimensions may have an impact on the motivation to communicate, the PDI has an observable influence on communication itself. Therefore, the PDI may be especially helpful in understanding a company's response to the shift from on-site audits to remote audits, based on the company's location.

PDI is defined as "the extent to which the less powerful members of organizations and institutions (like the family) accept and expect that power is distributed unequally" ***(2)***. This dimension considers the perception of inequality among the "followers," or the lower strata. Higher placement on the index indicates that hierarchy is clearly established and executed in society, without doubt or reason. Lower placement on the index signifies that people question authority and attempt to distribute power.

In considering the use of the PDI for the purposes of communications in remote auditing, the following generalities might be considered:

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- **High-PDI Societies:** In the hierarchy of authority, the auditor is considered above the auditee. The communication from the auditee will reflect the auditee's acceptance of this inequality.

- **Low-PDI Societies:** The auditor is doing his/her job and the auditee is doing his/her job. The auditee perceives an equal distribution of power, and the communication from the auditee will reflect this attitude. While the auditor has the authority to make requests, an auditee in a low-PDI society might respond to the auditor's requests in a professional manner that reflects their perception of the auditor as an equal.

Western European and Nordic countries like Denmark, Germany, Switzerland, and the United Kingdom, generally have a low PDI, although a few exceptions within Europe have mid-range or high PDIs, including France and Belgium. Eastern-hemisphere countries, like China and India, have high PDIs, and the United States lies in between *(3)*. **Table AII.1-1** provides an overview of location-specific PDIs from the Hofstede Insights website.

**Table AII.1-1** Geography-Specific PDI Examples

| High PDI | Middle PDI | Low PDI |
|---|---|---|
| South Asia | United States | Most of Central and Western Europe |
| Southeast Asia | Most of South America | Nordic Countries |
| Middle East | | Australia and New Zealand |
| Eastern Europe | | |
| Brazil | | |

### *Applying the Power Distance Index in Preparing for a Remote Audit*

Several caveats must be considered before applying Hofstede's PDI theory to cross-cultural communication:

- No organization should be stereotyped based on geography. An organization may have a PDI very different from the national trend.
- It may be tempting to oversimplify or to make moral inferences about behaviors exhibited by people in different regions. This would be contrary to the intended application of the PDI.
- Hofstede's cross-culture communication theory requires a detailed assessment of all six parameters, not only the PDI.
- Most cultural and communication measurement tools are 60% to 80% accurate. The overarching trend is the important aspect, not the individual value.

With these caveats in mind, the PDI can be a useful tool in creating a verbal communication plan in preparation for a remote audit. A two-step process can be used to sensitize teams.

**Step 1:** Recognize that there could be a cross-cultural variation in the PDI between the auditor and auditee organizations based on communication. If possible, determine the relevant PDIs that will be involved in an audit by conducting a generic paper-based review (survey). This can be used to help shape understanding of the expected communication patterns during the audit.

Identify the cross-cultural variation points and take steps to bridge the differences.

- **Communication from a High-PDI Auditee:** In a culture with a high PDI, the auditor is considered to be at a higher level than the auditee in the hierarchy of authority. Any communication from the auditee will reflect this perceived inequality in the relationship. For example, the inability to meet timelines is likely to be communicated in the form of a suggestion and accompanied with a statement of conformity such as: "We are working on uploading the required documents for you and are very hopeful to complete the task by the end of this week. Perhaps a little more time will help us to organize things better."

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- **Communication from a Low-PDI Auditee:** In a culture with a low PDI, the auditor and auditee are considered to be performing their respective functions, and the communication from the auditee will reflect equality in the relationship. For example, the inability to meet timelines would be communicated directly through such language as, "We foresee the task of uploading the documents extending beyond this week. We will need an additional two days."

- **Communication from a High-PDI Auditor:** The auditor's communication will reflect concentrated decision-making, stating expectations and not providing an opportunity for the auditee to express their opinion. The precise details of what the auditee needs to do will be the overtone of the communication. For instance, "It is important for me to review the documents tomorrow morning (my clock time). Upload them by this evening."

- **Communication from a Low-PDI Auditor:** The auditor's communication will reflect an intent to engage with the team of auditees and seek their agreement. For instance, "If we have to review the documents tomorrow morning (my clock time), when do you suggest you could have the documents uploaded?"

**Step 2:** Devise a communication strategy for the remote audit to counterweigh the challenges in communication based on the identified PDI of the organization or team of auditors/auditees *(4)*. **Table AII.1-2** provides suggestions and examples.

**Table AII.1-2** Communication Strategy Suggestions based on PDI of the Auditee

| Communication Aspects | High-PDI Auditee | Low-PDI Auditee |
|---|---|---|
| Ambiguity | The team of auditees would expect the team of auditors to provide clear directions, with stepwise instructions to follow, e.g., "Every day we will set aside 30 minutes of the 4 hours for planning and review." | Collaboration is a paramount aspect and will be reflected in the communication. Seeking suggestions for planning and executing a remote audit is an expectation, e.g., "How do you suggest we plan for our next day's review? Will 30 minutes be suitable for your team?" |
| Communication Strategy for Auditors | Provide clear steps to be followed by the auditee; asking for suggestions may be confusing for the high-PDI auditee. | Seek consensus for action items. Telling a low-PDI auditee what to do may be perceived as overstepping. |
| Communication Strategy for Auditees | Seek clarity and steps to be followed. | Provide recommendations or solutions. |
| Reasoning | Communications should include detailed instruction on what is required and how it is to be provided, e.g., "A summary report is required along with the investigation document. The report needs to include a table of contents. The summary need not be more than three pages in length." | Communications should illustrate facts and not mere instructions. That is, the reason behind acting or not acting needs to accompany the statements made, e.g., "The investigation document you have shared is more than 50 pages. Will it be possible to send a table of contents and a summary to expedite the review?" |
| Communication Strategy for Auditors | Provide the auditee with clear details of specific requirements and expectations. | Engage with the auditee in explaining what is sought and why it is sought in the requested form/format. |
| Communication Strategy for Auditees | Seek background information and steps to be followed. | Provide recommendations or solutions and options backed with reasons. |
| Accountability | First- and second-person communication is recommended to establish accountability, e.g., "We expect you will check with your IT department and confirm the feasibility of using the e-sharing platform for document exchange by next Monday." | Third-person communication is preferable for business communication, e.g., "Early confirmation of the document-sharing platform and its security precautions would be appreciated." |
| Communication Strategy for Auditors | Provide clear details of specific responsibilities. | Engage with the team and propose action items for consensus. |
| Communication Strategy for Auditees | Seek clarity on whom to contact and subsequent steps to be followed. | Provide recommendations or solutions and options backed with reasons. |


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### *Examples of the Potential Application of PDI*

PDA conducted a survey that can be used to help understand whether culture impacts the communications between the auditor and the auditee during remote audits based on the geographic location of the respondent, that is, high or low PDI *(5)*. While the survey questions specifically addressed regulatory inspections rather than audits, cultural differences are likely to be as relevant in external audits of suppliers and CMOs as in inspections.

- The purpose of this analysis is to sensitize the reader to differences in the communication based on cultural backgrounds. The differences shown in the survey results may not be statistically significant and are presented for the purpose of illustration only.
- This analysis includes only certain questions from the survey that highlight cultural and communication nuances.

The survey was open to global respondents, who were given the option of selecting their region as North America, Europe, Asia-Pacific, Middle East, or Africa. Responses from Africa and the Middle East were not included in this analysis due to the low number of respondents from those areas.

This analysis presents each of the four selected questions, followed by a table of responses and a figure representing those responses.

### Question 1

Would you agree to make employees available for remote (audio and/or video) interviews by the investigator?

**Table AII.1-3** Response to Question about Remote Interviews by Geography

| | Yes, All Requested Employees | Yes, but Only Specific Employees | Depends on Regulator's Legal Authority | Not Sure | Other |
|---|---|---|---|---|---|
| North America | 9 (47.4%) | 5 (26.3%) | 2 (10.5%) | 2 (10.5%) | 1 (5.3%) |
| Europe | 8 (53.3%) | 6 (40.0%) | 0 (0.0%) | 0 (0.0%) | 1 (6.7%) |
| Asia-Pacific | 5 (100.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) |
| | | | | | ***Answered 39*** |

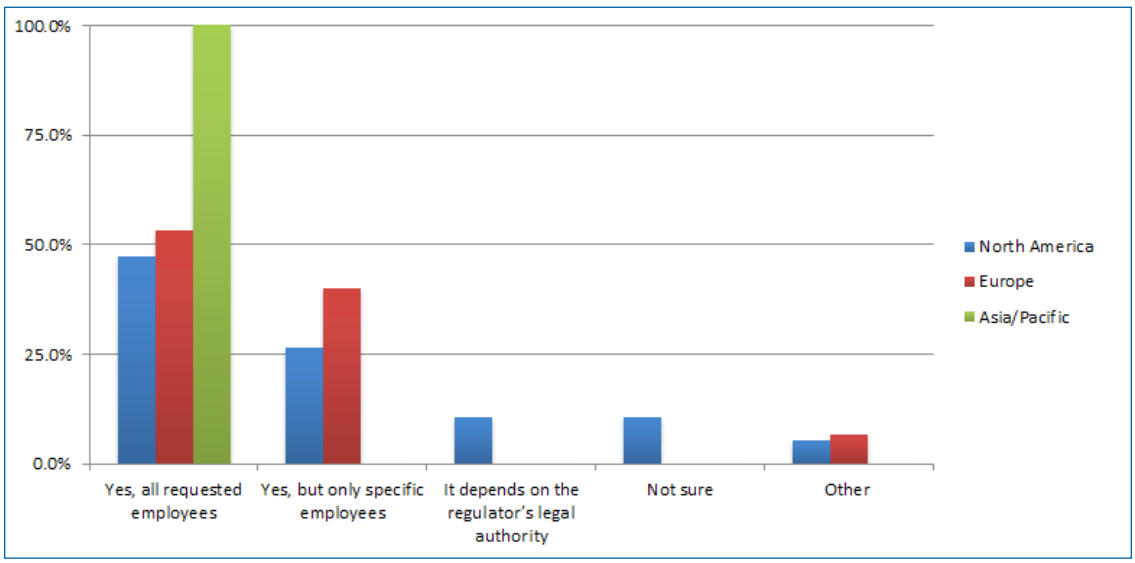


**Figure AII.1-1** Response to Question about Remote Interviews by Geography


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In response to this question, the respondents from high-PDI geographic areas appear to show an acceptance of the hierarchy of inspector versus inspected site in agreeing to allow all requested employees to take part in remote interviews. The respondents from low-PDI geographic areas opted more frequently for other options, such as providing only specific employees, indicating a willingness to seek to distribute power more equally.

**Question 2**

If asked during a remote inspection, would you allow an inspector to view your own live video feed of your facility, for instance, by viewing your [closed circuit television (CCTV)] footage or other internal camera feed?

**Table AII.1-4** Response to Question about Live Video Feed by Geography

| | Yes, Inspector Could Access Directly | Yes, Staff would Access and Share Images/Video with Inspector | Depends on Regulator and their Authority; Might Allow Some Regulators, Not All | No | Not Sure |
|---|---|---|---|---|---|
| North America | 0 (0.0%) | 7 (38.9%) | 6 (33.3%) | 3 (16.7%) | 2 (11.1%) |
| Europe | 1 (6.7%) | 4 (26.7%) | 4 (26.7%) | 4 (26.7%) | 2 (13.3%) |
| Asia-Pacific | 2 (50.0%) | 1 (25.0%) | 1 (25.0%) | 0 (0.0%) | 0 (0.0%) |
| | | | | | ***Answered 37*** |

Respondents from low-PDI geographic locations would refuse access to live video feeds, suggesting a willingness to distribute power. In contrast, none of the respondents from high-PDI geographic locations would refuse access to a live video feed, indicating acceptance of the hierarchy of inspector versus inspected site.

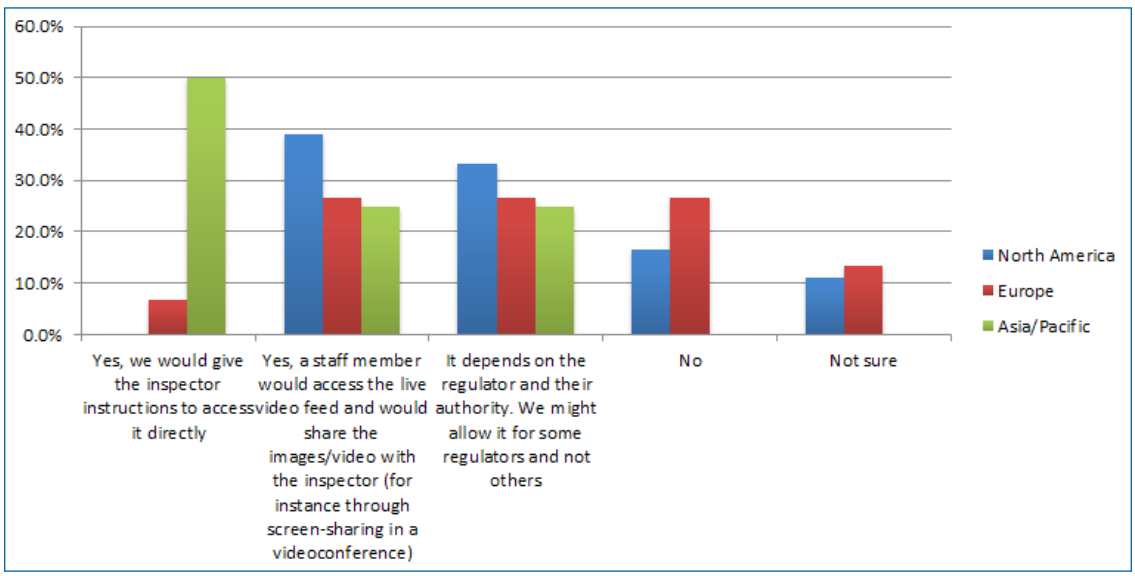


**Figure AII.1-1** Response to Question about Remote Interviews by Geography


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## Question 3

During a remote inspection conducted by videoconference (e.g., Zoom, Microsoft Teams, GoToMeeting), would you allow the regulatory authority to record the proceedings?

**Table AII.1-5** Response to Question about Recording Proceedings by Geography

| | Yes, Would Agree without Limitation | Yes, if Inspected Site hosts Meeting on Controlled Platform Approved by IT | No | Not Sure | Other |
|---|---|---|---|---|---|
| North America | 2 (11.1%) | 5 (27.8%) | 2 (11.1%) | 8 (44.4%) | 1 (5.6%) |
| Europe | 0 (0.0%) | 3 (20.0%) | 6 (40.0%) | 3 (20.0%) | 3 (20.0%) |
| Asia-Pacific | 1 (25.0%) | 2 (50.0%) | 0 (0.0%) | 1 (25.0%) | 0 (0.0%) |
| | | | | | ***Answered 37*** |

As with **Question 2**, some respondents from low-PDI locations indicated their willingness to refuse to allow recording of inspectional proceedings, which may indicate an interest in sharing power. None of the respondents from high-PDI locations indicated that they would refuse the recording, indicating acceptance of the hierarchy of inspector versus inspected site.

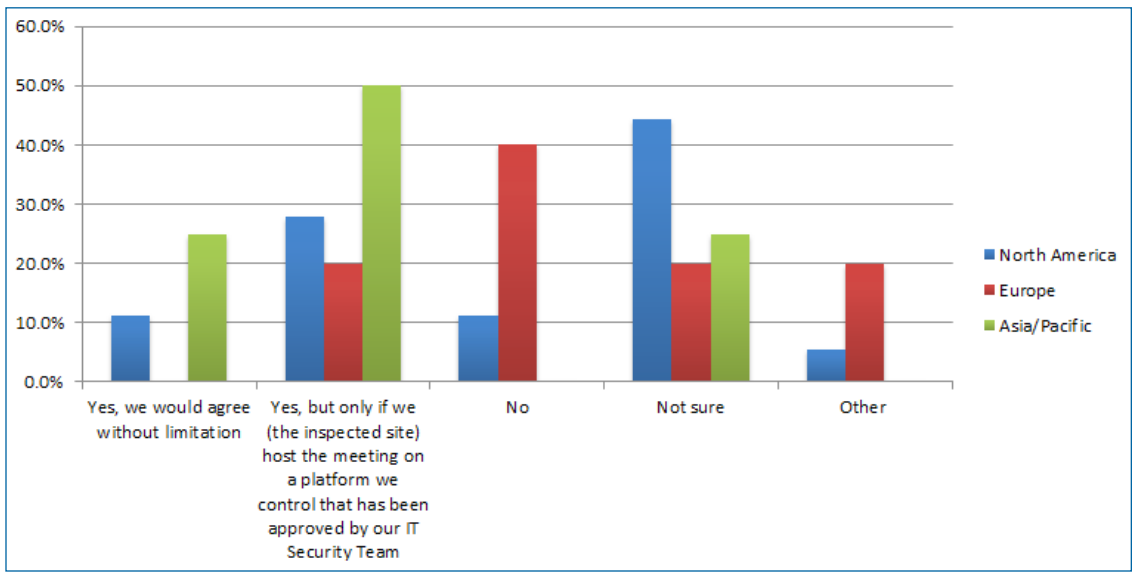


**Figure AII.1-3** Response to Question about Recording Proceedings by Geography


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## Question 4

If the U.S. FDA notified you that they wanted to conduct a "voluntary regulatory assessment," which would not technically be an inspection, but the results of which could determine approval or withholding of approval of a pending submission (e.g., NDA, ANDA, BLA), would you (1) agree to participate, (2) refuse to participate in this voluntary assessment, but state that you would fully cooperate with an official inspection that met statutory requirements, (3) not sure, or (4) other.

**Table AII.1-6** Responses to Question about FDA Voluntary Regulatory Assessment by Geography

| | Agree to Participate | Refuse to Participate in Voluntary Assessment, but Would Cooperate with Official Inspection | Not Sure | Other |
|---|---|---|---|---|
| North America | 8 (47.1%) | 3 (17.6%) | 5 (29.4%) | 1 (5.9%) |
| Europe | 8 (61.5%) | 0 (0.0%) | 2 (15.4%) | 3 (23.1%) |
| Asia-Pacific | 4 (100.0%) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) |
| | | | | ***Answered 34*** |

In response to this question, the respondents from high-PDI areas appear to have accepted the hierarchy of inspector versus the inspected, in that they have not opted to refuse the assessment. In contrast, respondents from low-PDI areas would attempt to distribute the power more equally.

## Rationale

To assure a successful audit, it is crucial that all participating individuals be sensitized to communication differences as part of the audit preparations. Advancing efforts to improve communication and understanding between the parties involved will make for a more robust, efficient, and clear information exchange.

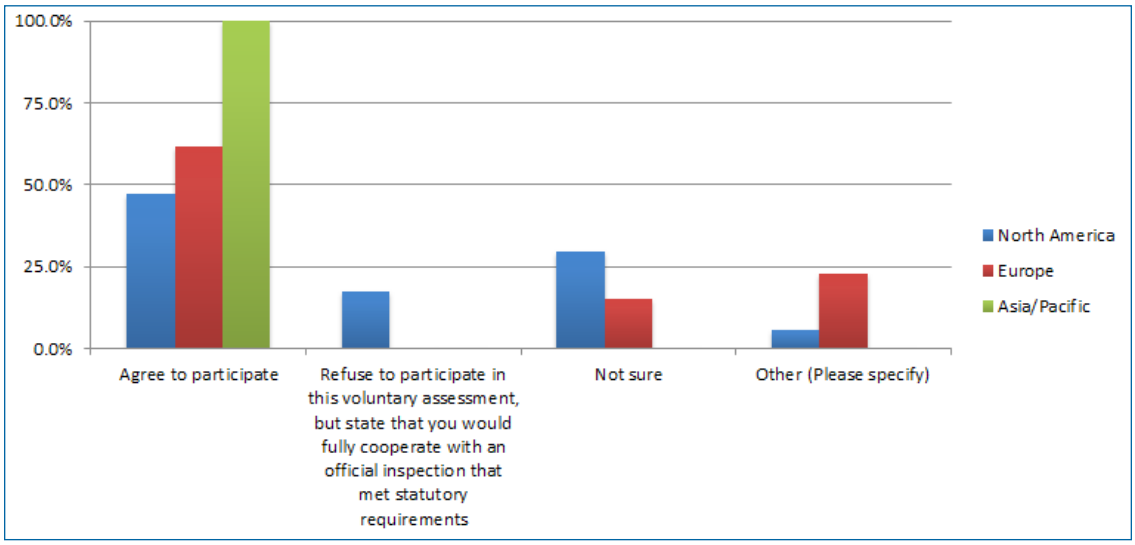


**Figure AII.1-4** Response to Question about FDA Voluntary Regulatory Assessment by Geography


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# Points to Consider in Remote and Hybrid GMP/GDP Audits

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Bethesda, MD 20814 USA
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